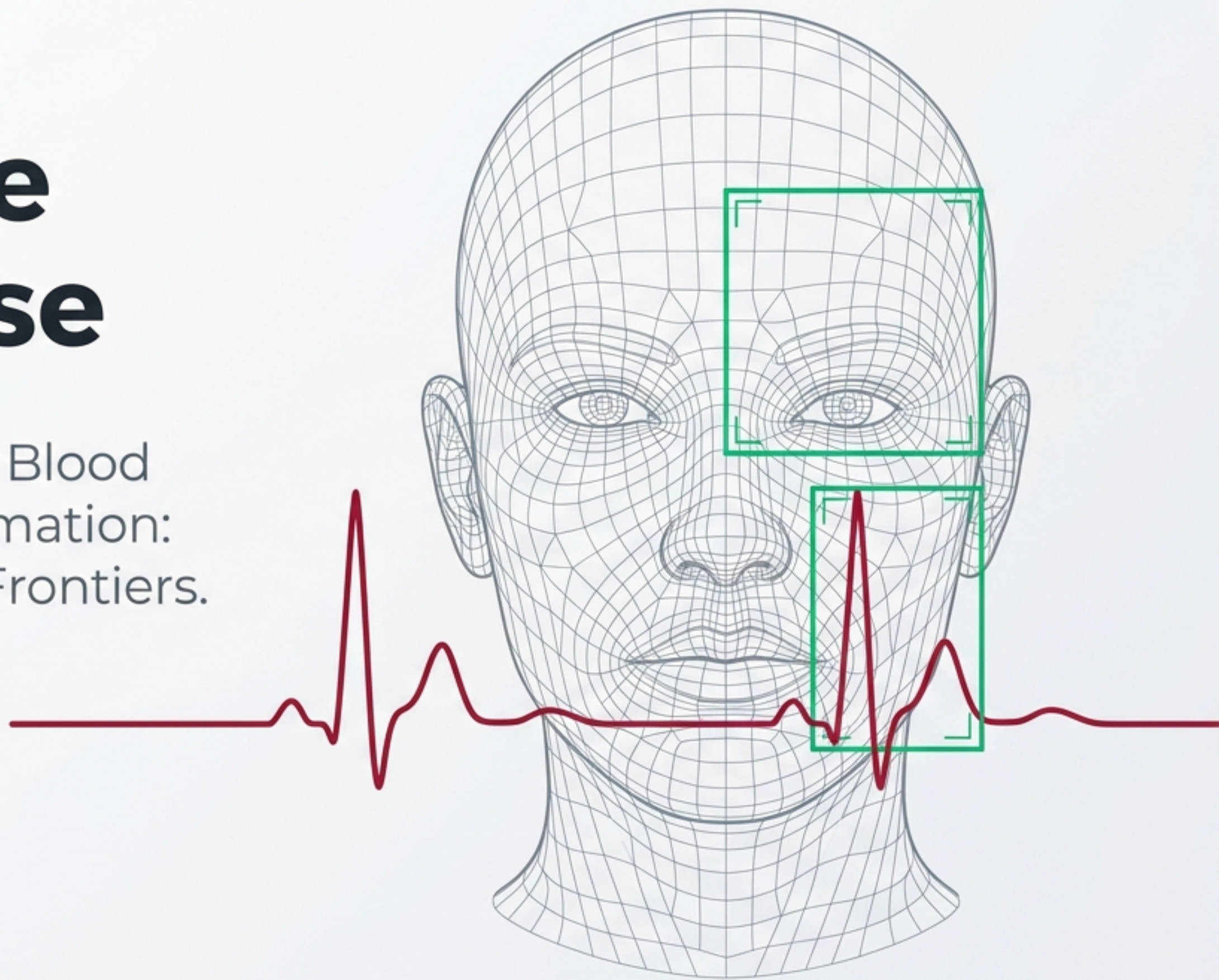
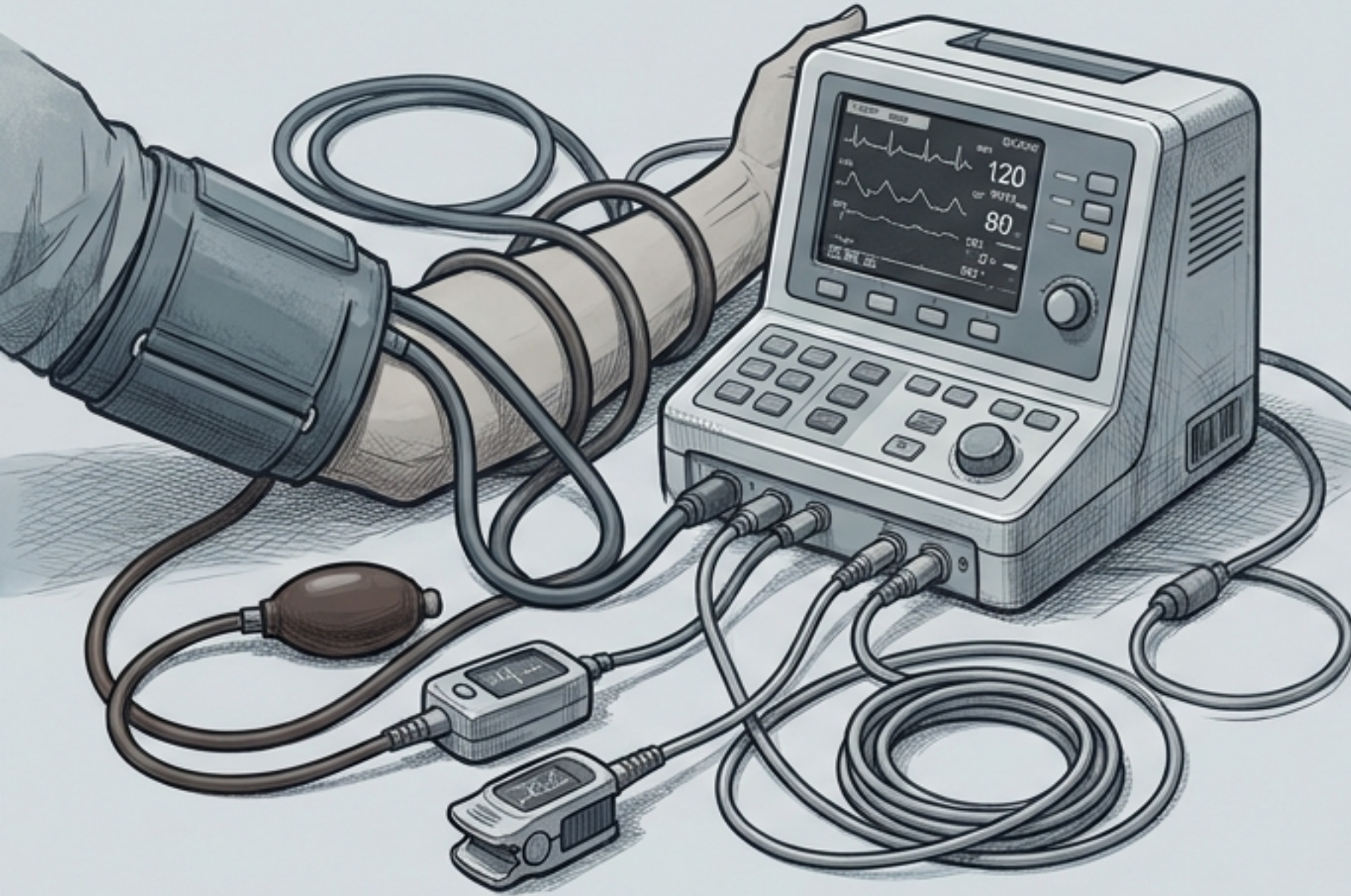


Decoding the Invisible Pulse

Smartphone-Based rPPG for Blood Pressure and Vital Signs Estimation: State-of-the-Art and Future Frontiers.



Ambient Intelligence for 1.4 Billion



The Clinical Burden

Hypertension affects 1.4 billion adults globally (WHO, 2024). Traditional episodic cuff readings fail to capture masked or white-coat hypertension.

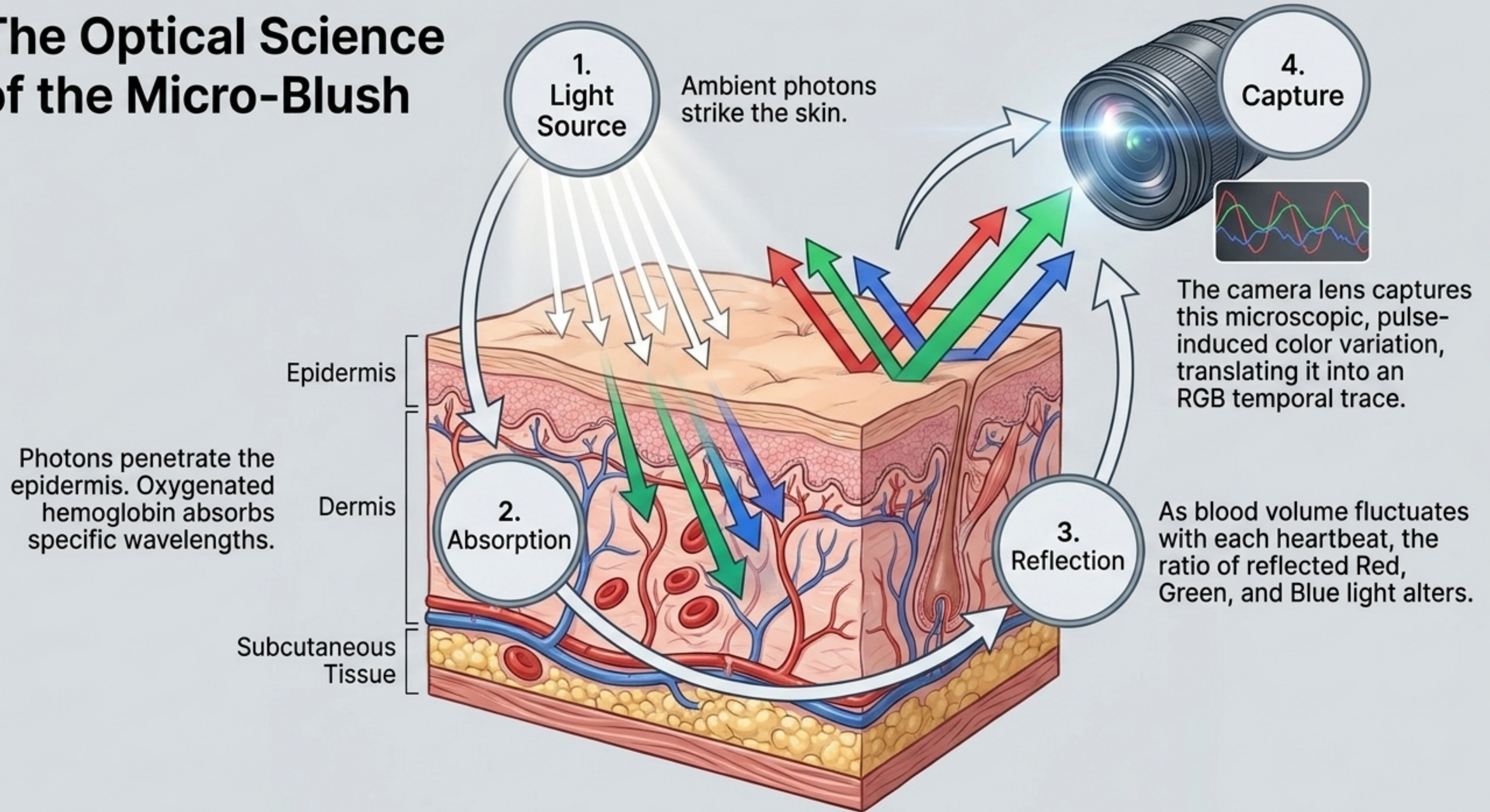
The Hardware Bottleneck

Wearables and cuffs suffer from poor long-term compliance, physical degradation, loss of contact, and skin irritation in fragile patients.

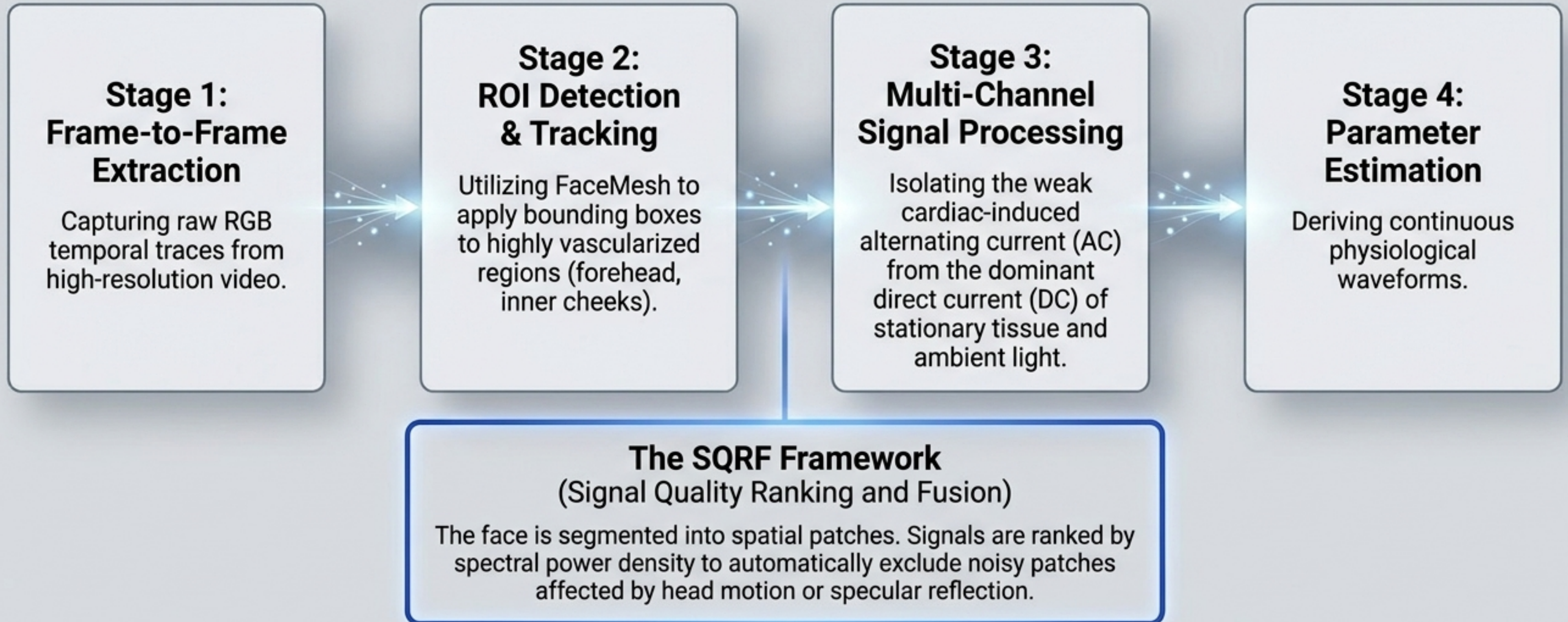
The Software Solution

Remote Photoplethysmography (rPPG) transforms the ubiquitous smartphone camera into a continuous, clinical-grade diagnostic tool via ambient light analysis.

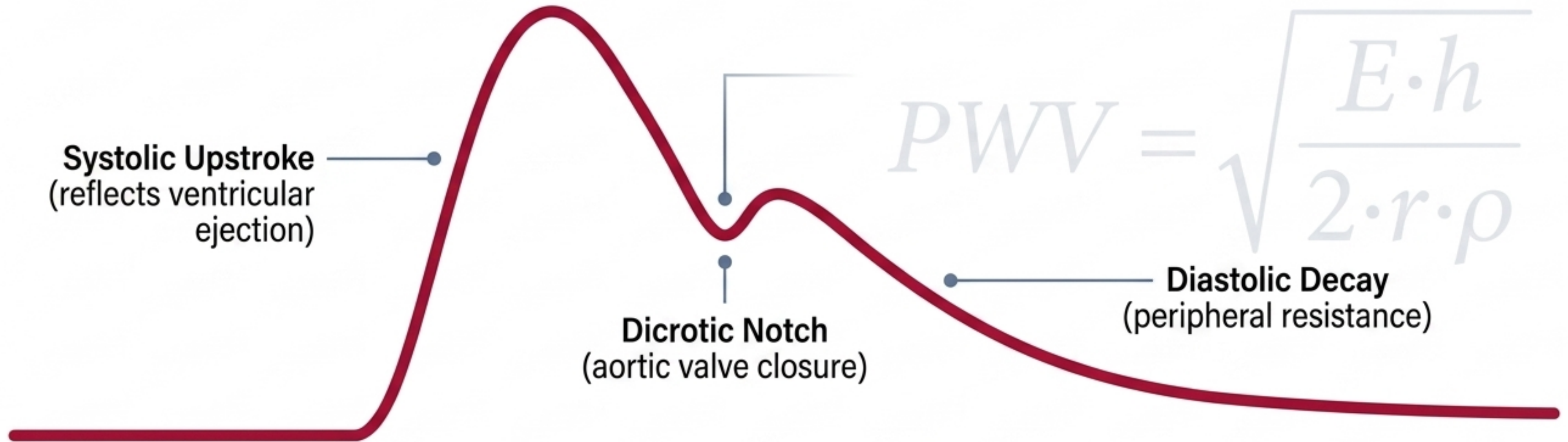
The Optical Science of the Micro-Blush



Pixels to Pulse: The Extraction Architecture



Extracting Blood Pressure from Waveform Morphology



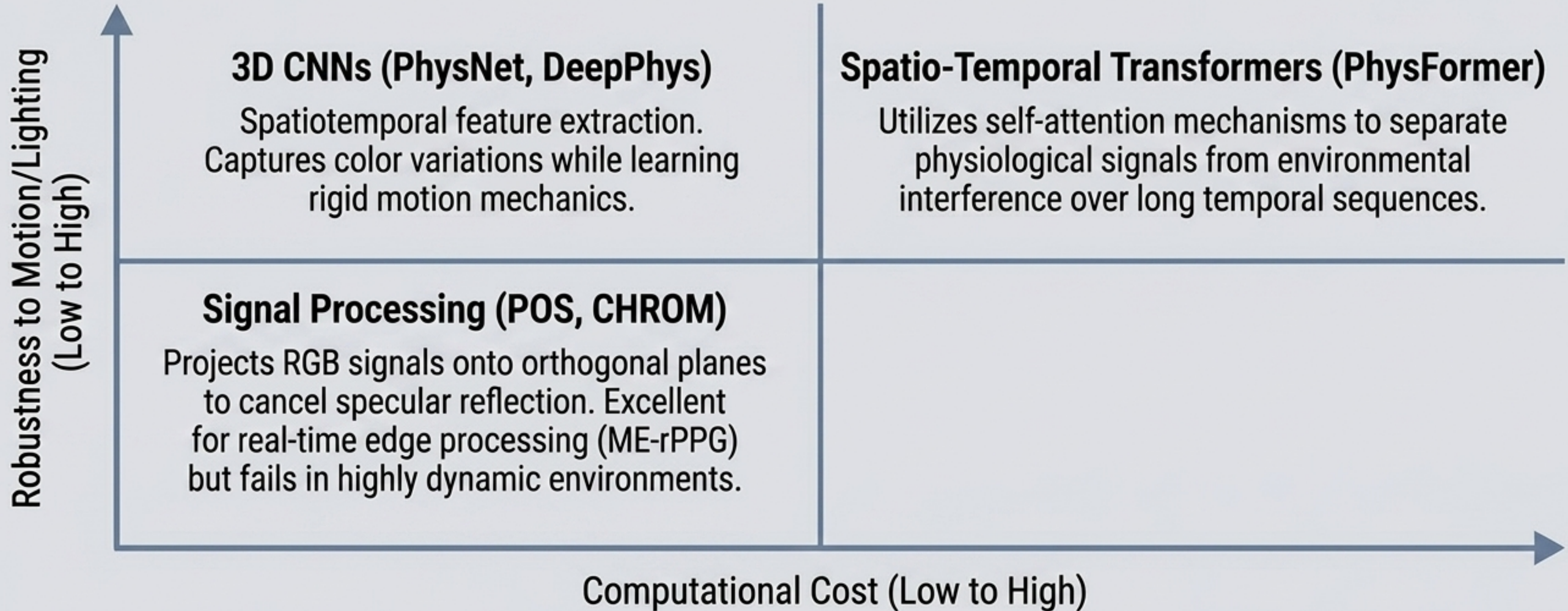
Pillar A: Morphological Analysis

Heart rate is merely peak-to-peak frequency. BP requires preserving high-fidelity biophysical features that reflect systemic vascular compliance.

Pillar B: Pulse Wave Velocity (PWV)

Utilizing phase differences between distinct facial ROIs (e.g., forehead to cheek) to approximate Pulse Transit Time (PTT) without a physical cuff.

Mapping the Analytical Landscape



The Goal: Pushing high-robustness Transformer architectures toward the low-compute axis for scalable edge-device deployment.

Benchmarking the Algorithms: The Dataset Evolution

Dataset	Cohort Size	Environmental Variables	Reference Ground-Truth
PURE (2014)	10 subjects	Controlled lab lighting, rigid motion	Finger pulse oximetry
UBFC-rPPG (2019)	42 subjects	Stationary webcam	Continuous PPG, HR
MMPD (2023)	33 subjects	Mobile phone, 4 lighting setups, walking/talking	Continuous transmissive PPG
CLBP-300 (2024)	300 subjects	iPhone/DSLR, Indoor/Outdoor (Lux tracked)	SBP/DBP (Omron/Rossmax cuffs)
MCD-rPPG (2024)	600 subjects	3 camera angles, resting vs. post-exercise	13 biomarkers including 100 Hz medical PPG, SBP/DBP, ECG

Clinical Validation: The ISO 81060-2 Mandate

The Gold Standard

Dual Auscultation by two independent observers (99.8% agreement within ≤ 4 mmHg).

The Cohort Requirement

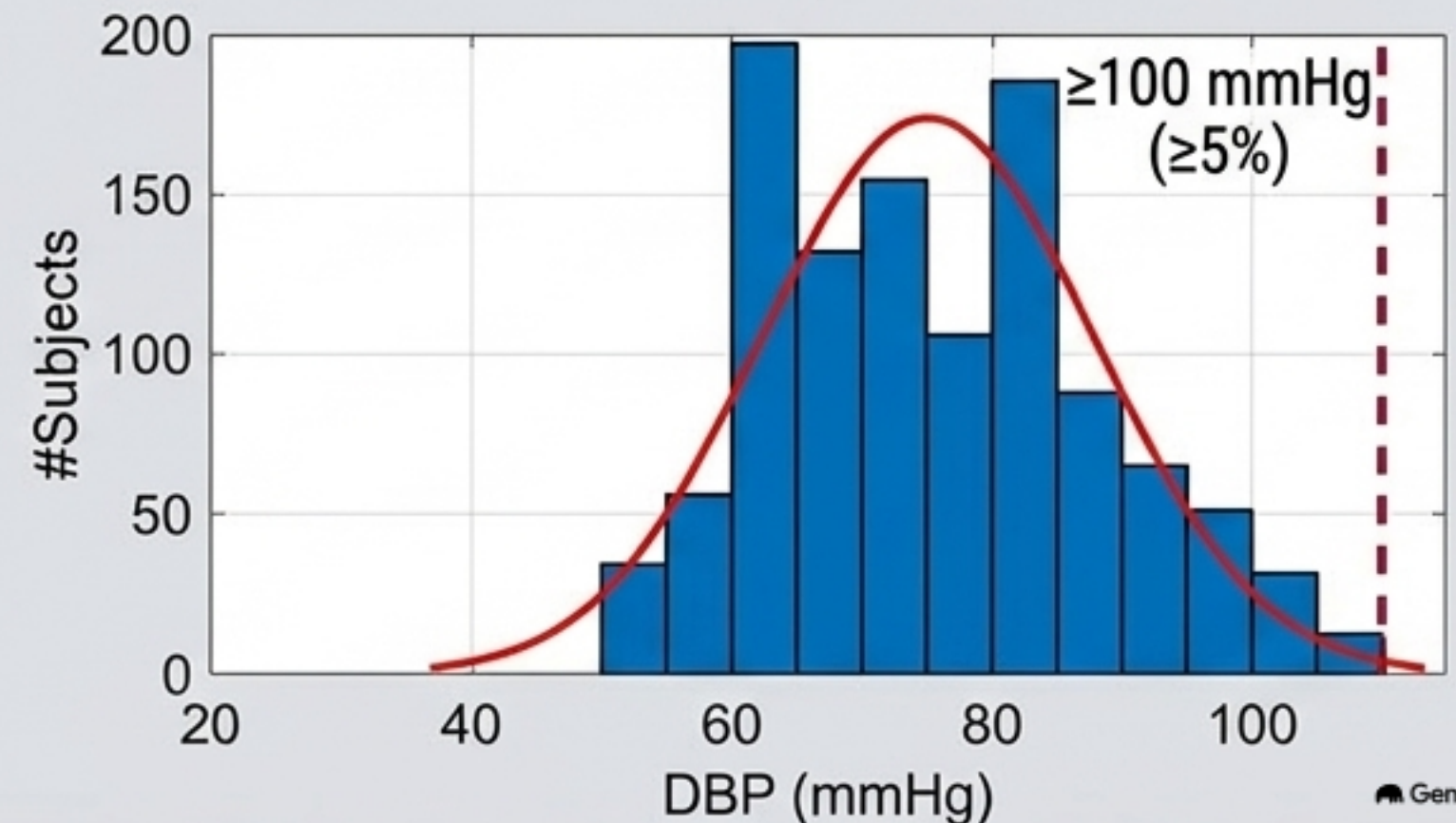
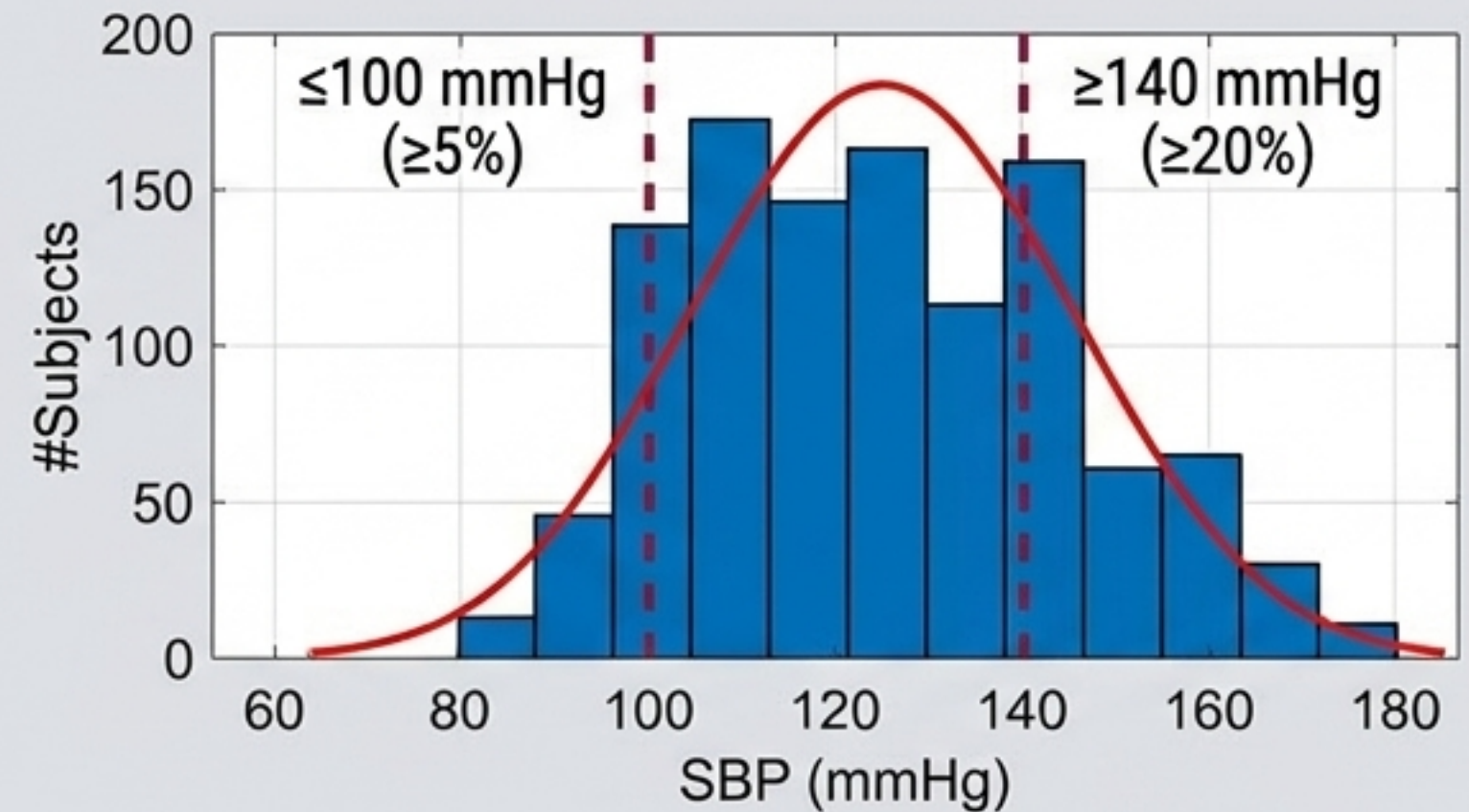
Minimum 85 valid subjects with strict demographic and hemodynamic distributions.

The Distribution Challenge

Models cannot just be accurate on healthy patients. The standard demands:

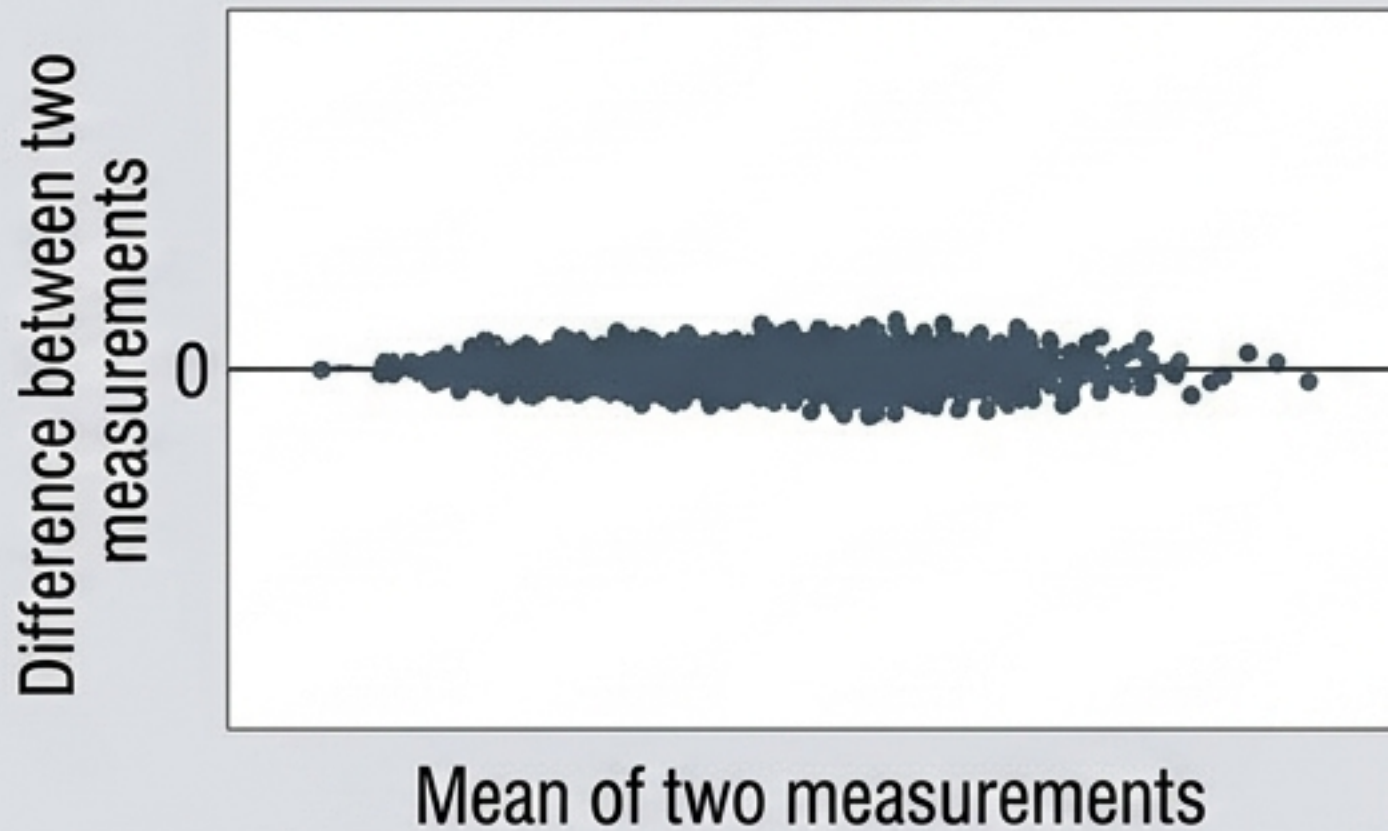
- $\geq 5\%$ of readings with SBP ≤ 100 mmHg
- $\geq 20\%$ of readings with SBP ≥ 140 mmHg
- $\geq 5\%$ of readings with DBP ≥ 100 mmHg

The SGH Preoperative Study Case: Large trials (e.g., Singapore General Hospital, 200 subjects) easily meet SBP requirements (26% ≥ 140 mmHg), but frequently fail DBP distribution requirements (0% ≥ 100 mmHg), invalidating clinical certification.



Current Clinical Efficacy and Volatility

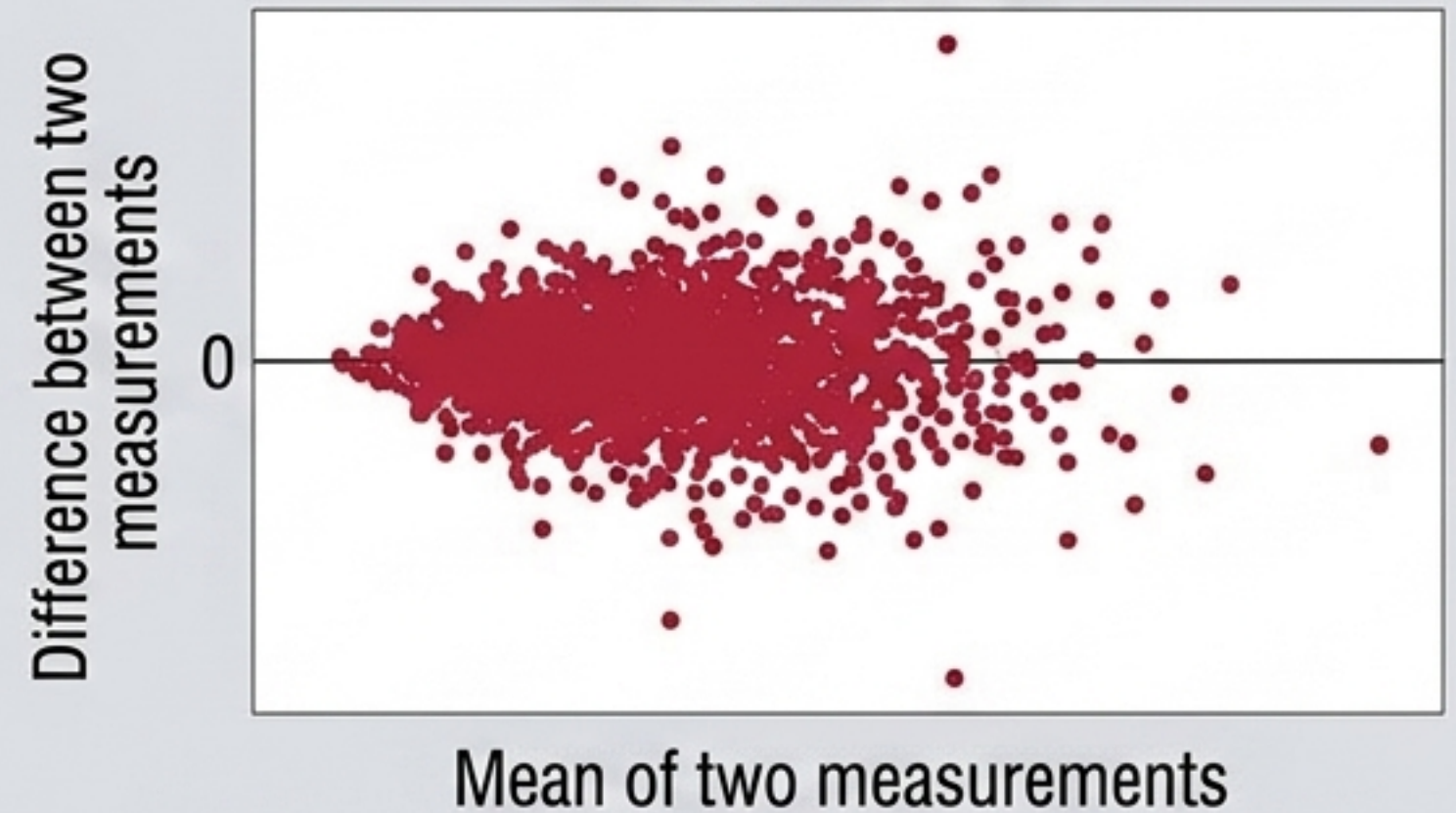
Heart Rate



Heart Rate & SpO2 (Maturity: High)

- Commercial solutions (e.g., Lifelight, Binah.ai) achieve highly accurate results with <1 BPM mean error. FDA/CE-track viable.

Blood Pressure



Blood Pressure (Maturity: Evolving)

- Promising, but suffers from high standard deviations.
 - Lifelight Trial (8,500 subjects): Achieved an average 2-3 mmHg error for Systolic BP, but exhibited a massive standard deviation of nearly ± 15 mmHg.
 - Binah Trial (120 subjects): Achieved 6.4 mmHg error with a ± 4.8 mmHg standard deviation, but strictly required preset stands and idealized distances, eliminating 3-SD outliers.

Barriers to Ubiquity



Illumination & Environmental Noise

Low Signal-to-Noise Ratio (SNR) in dim lighting; camera auto-exposure and white-balance adjustments actively destroy the underlying AC physiological signal.



Demographic Equity (Fitzpatrick Scale)

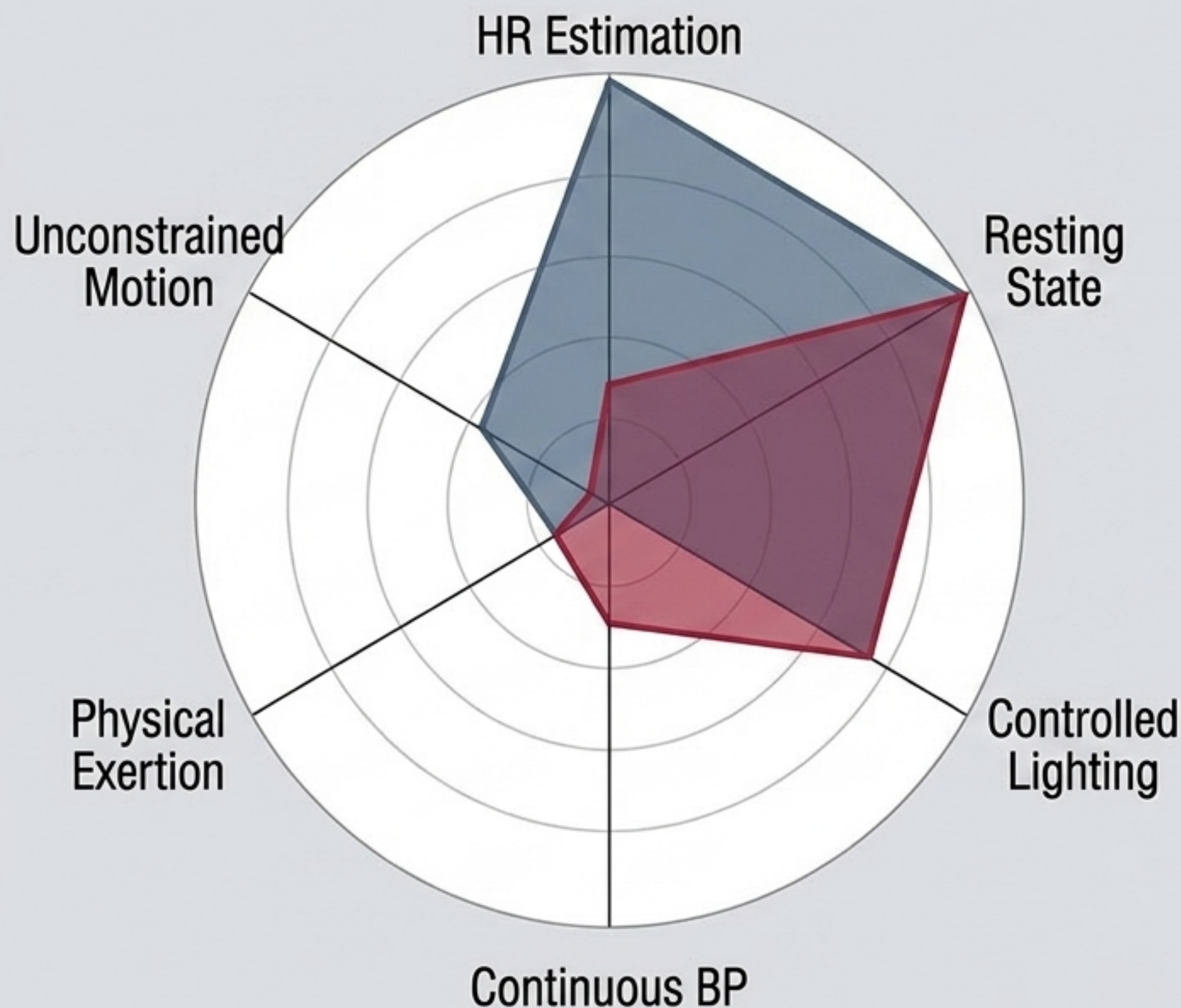
Melanin inherently absorbs light. Darker skin tones reflect less RGB data back to the sensor, significantly reducing the signal strength of the capillary bed and causing algorithmic bias if training data is unrepresentative.



Motion Artifacts

Talking, micro-expressions, and non-rigid head movements (or hand tremors holding the phone) easily drown out the microscopic color variations of the BVP.

The Missing Pieces in the rPPG Ecosystem



The Dynamic Data Gap

Over-reliance on synthetic data or resting-state measurements. Models fail catastrophically during physical exertion (hence the need for post-exercise datasets like MCD-rPPG).

The Confidence Gap

Absence of standardized "Confidence Scores" embedded into algorithms. Systems must know when they are failing and reject poor-quality measurements automatically.

The Morphological Gap

Most models regress BP from generic features rather than explicitly mapping localized phase-shifts and precise Pulse Transit Times.

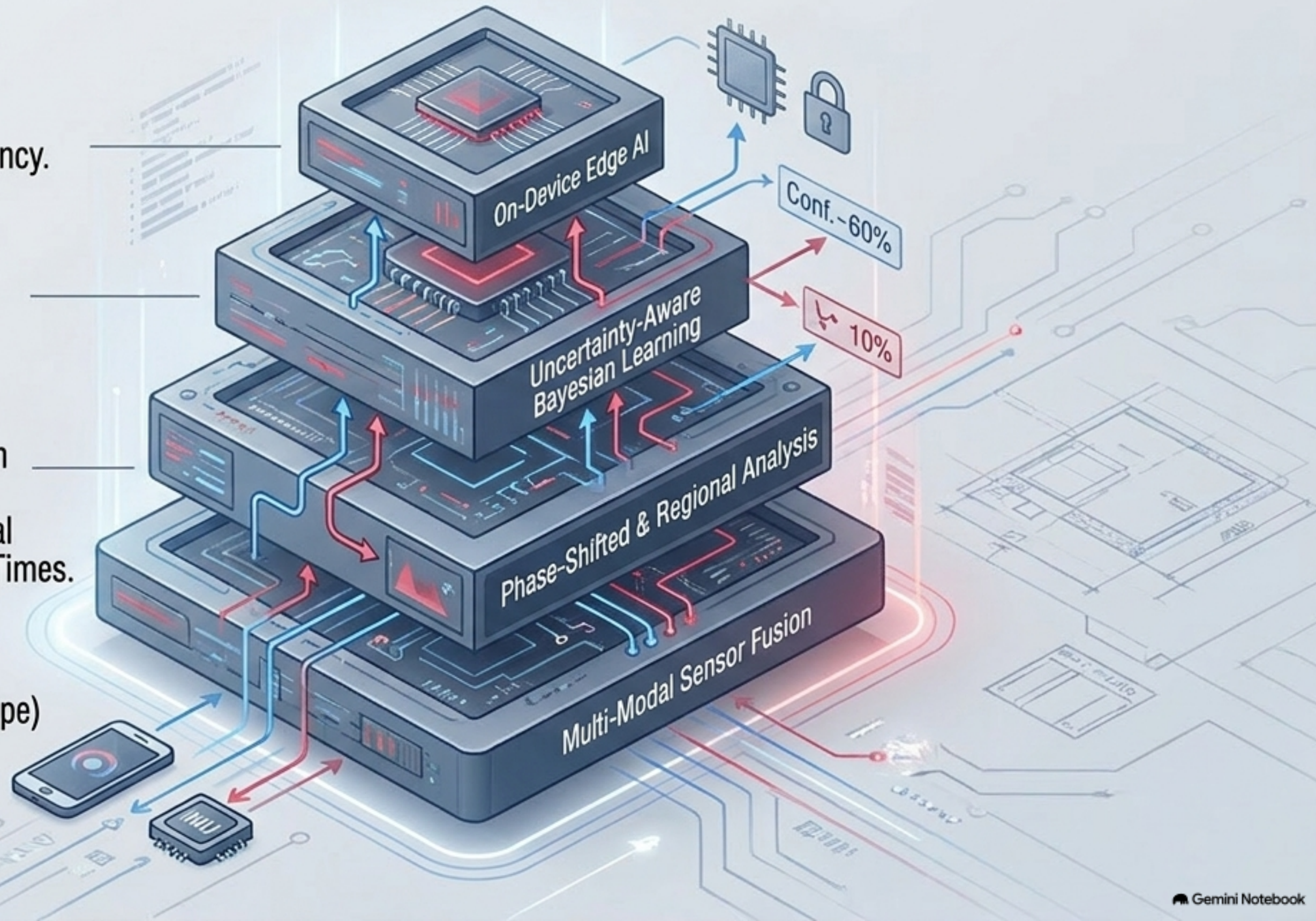
The Next Frontier: A Unified System Framework

Processing strictly on-device (via **state-space models like ME-rPPG**) to ensure zero-latency feedback and absolute biometric privacy without cloud dependency.

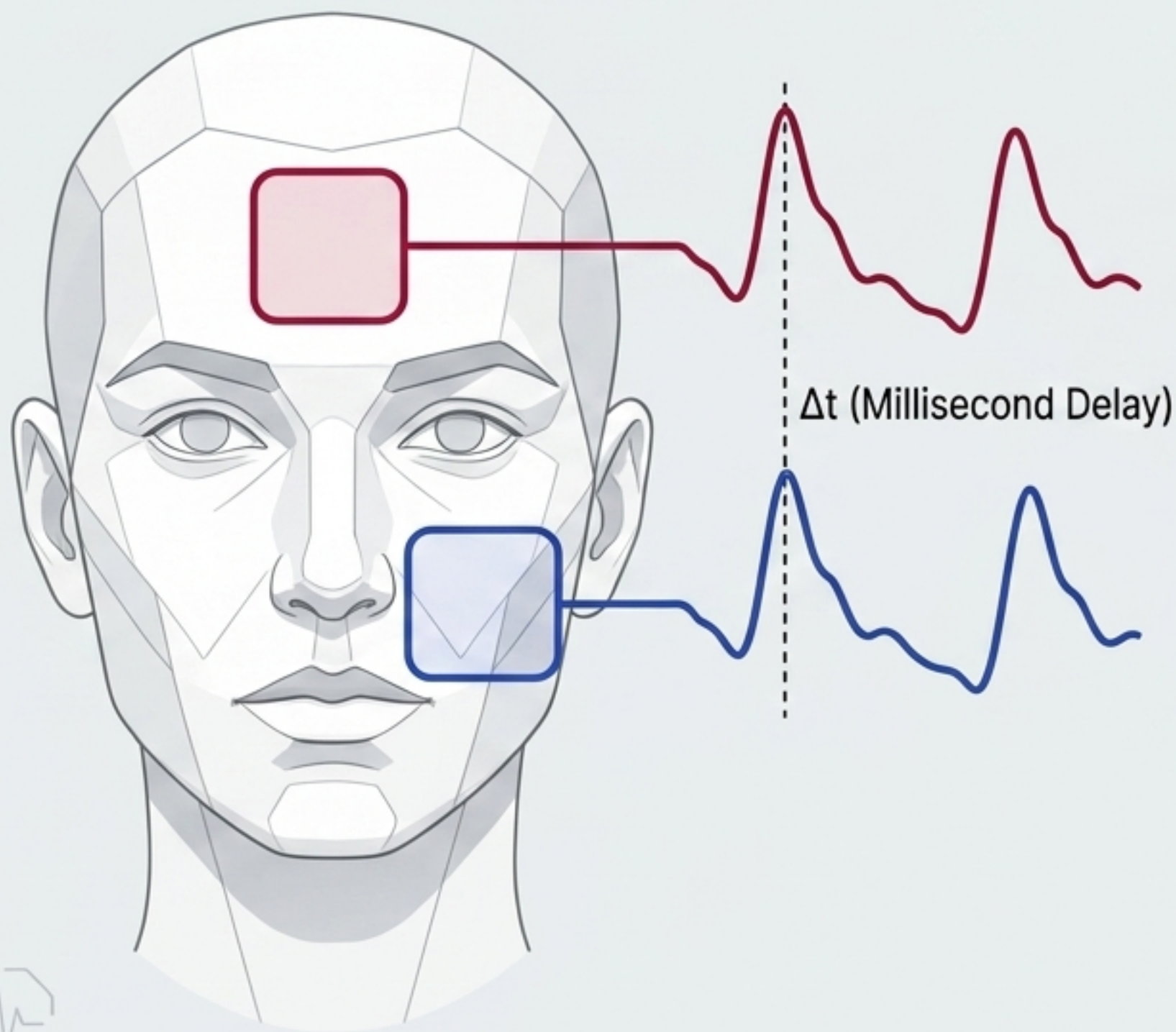
Deep learning models (BNNs) that don't just output a single BP number, but output a paired Confidence Score, enabling proactive user coaching.

Utilizing independent, dual-ROI extraction (e.g., forehead vs. cheek) via networks like DRP-Net to calculate explicit temporal discrepancies and precise Pulse Transit Times.

Combining camera-based rPPG with smartphone IMU (accelerometer/gyroscope) data to explicitly model and cancel out hand-tremors and device motion.



Mechanism Focus: Phase-Shifted rPPG



The Physics:

Blood travels from the heart to the facial arteries. Because the forehead and cheeks are at slightly different distances along the vascular tree, the pulse arrives at different milliseconds.

The Computation:

Advanced networks (like DRP-Net and BBP-Net) analyze this microscopic temporal discrepancy between the phase-shifted rPPG signals.

The Result:

This unlocks true Pulse Transit Time (PTT) analysis purely from video, bypassing the need for an external ECG reference and dramatically lowering the Mean Absolute Error for blood pressure predictions.

Mechanism Focus: Uncertainty-Aware AI



The Problem with Black Boxes:

Standard neural networks provide deterministic outputs even when fed garbage data (e.g., a heavily shadowed face, motion blur).

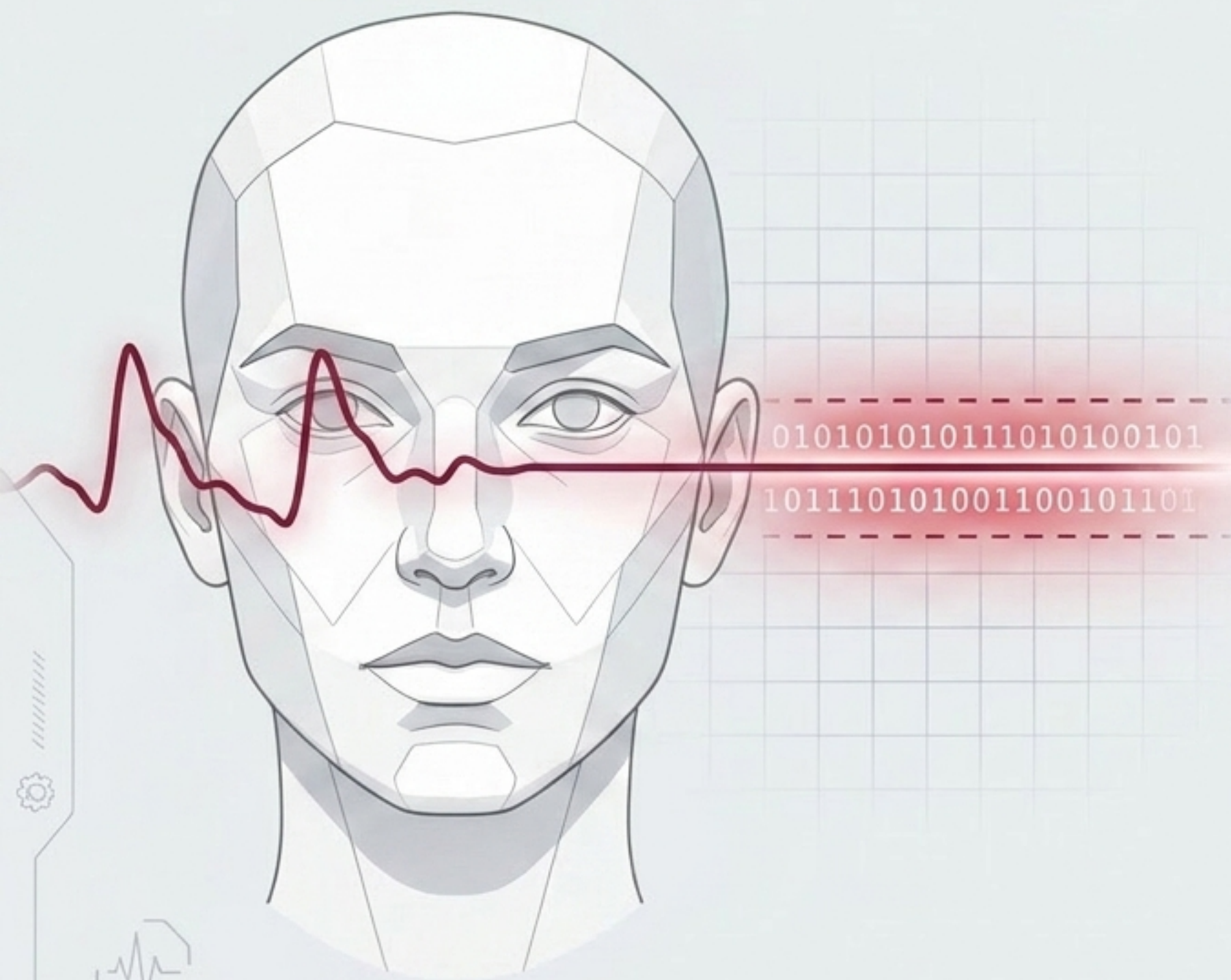
Bayesian Neural Networks (BNNs):

By implementing Bayesian ensemble learning (e.g., U-FaceBP), the network models its own uncertainty by analyzing the variance in the **signal distribution**.

Digital Coaching:

The system acts as a smart digital coach, **rejecting low-SNR signals** in real-time. This ensures the final clinical output is derived only from **high-fidelity, verified waveform morphology**, building trust with clinicians.

Making the Invisible Actionable



From Novelty to Utility:

rPPG has evolved past computer vision proofs-of-concept into a clinically viable pathway for continuous, cuffless cardiovascular monitoring.

The Data Imperative:

Achieving the ISO 81060-2 standard requires abandoning highly controlled lab datasets in favor of diverse, multi-modal registries (like MCD-rPPG).

The Final Paradigm:

The transition from rigid hardware sensors to ambient, software-driven intelligence will define the next era of proactive, democratized healthcare.